

Guy Paterson

Engineering Portfolio

Airbus

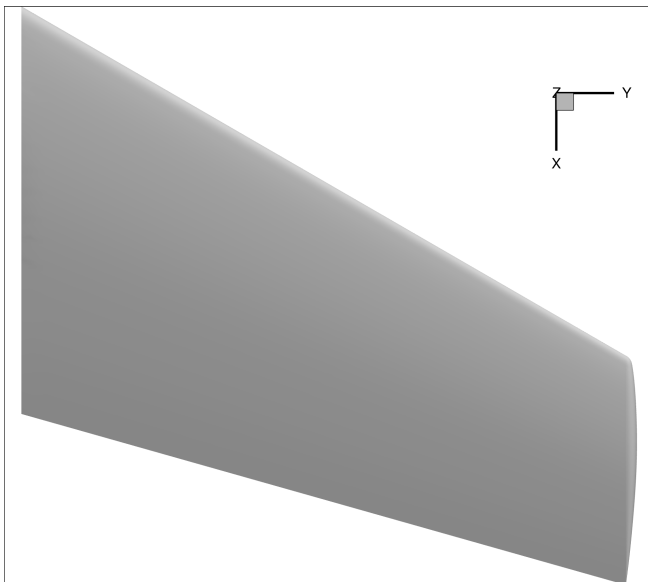
CFD Engineer | Flight Physics, Aerodynamics Modelling | 2025–2026

Most of my Airbus work involved proprietary aircraft geometries, including the A321 and A350, and therefore cannot be shared. The evidence below highlights exceptions: work conducted on open-source geometries.

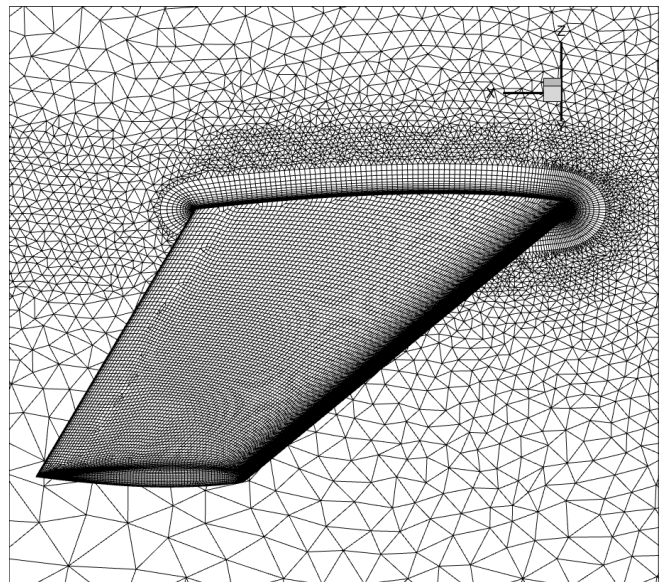
All results presented below were computed using [CODA](#), European aerospace's next-generation CFD code, developed by Airbus, DLR and ONERA. This work supported CODA's entry into service as it replaces the legacy [TAU](#) CFD code.

E1 ONERA M6

The ONERA M6 is a swept-wing geometry widely used to validate compressible CFD methods. The 1979 ONERA experiment produced extensive surface-pressure data across a range of Mach numbers and angles of incidence, providing a benchmark for assessing solver accuracy, especially for separated flows.



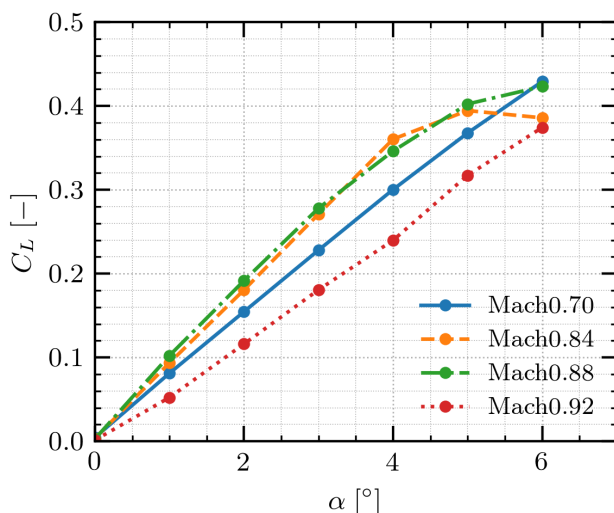
ONERA M6 geometry.



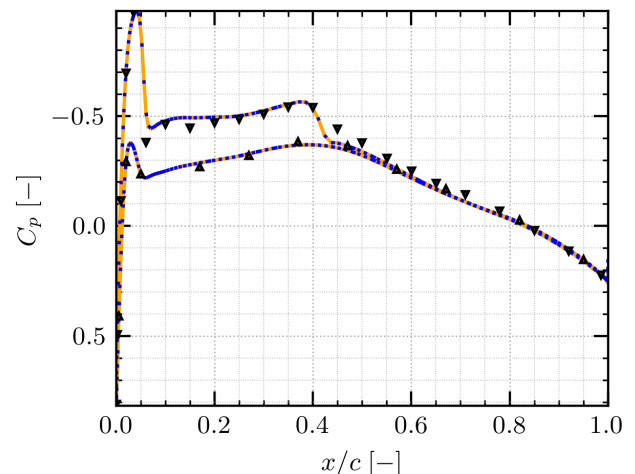
ONERA M6 mesh.

E1.1 Test Case Integration

One of my first tasks at Airbus was to digitise the ONERA M6 experimental data from archived PDFs to create a validation dataset for Airbus and integrate the test case into the automatic testing framework.



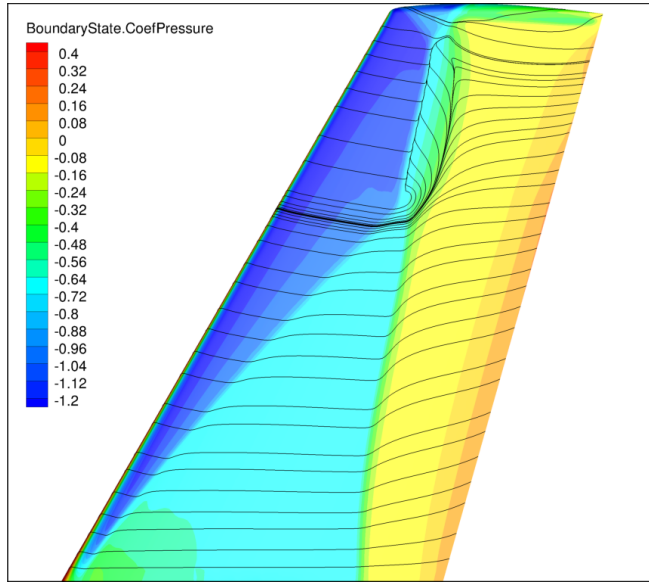
CFD results of the broad test matrix.



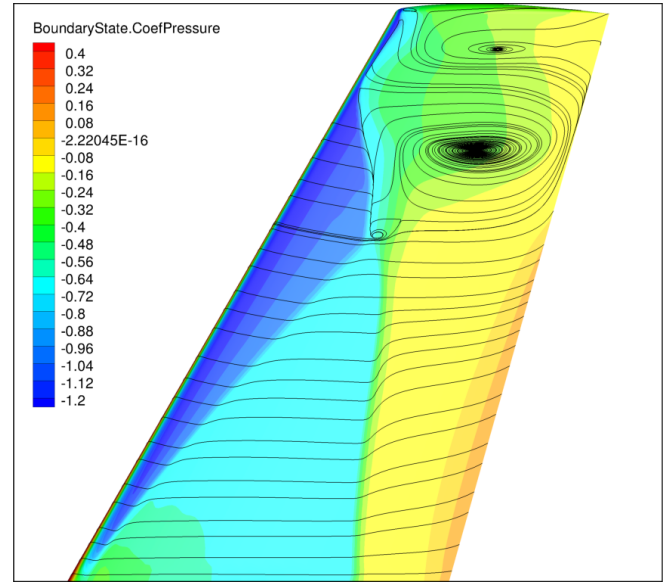
Version regression at Mach = 0.84, $\alpha = 3^\circ$, $\eta = 45\%$.

E1.2 RSM Investigation

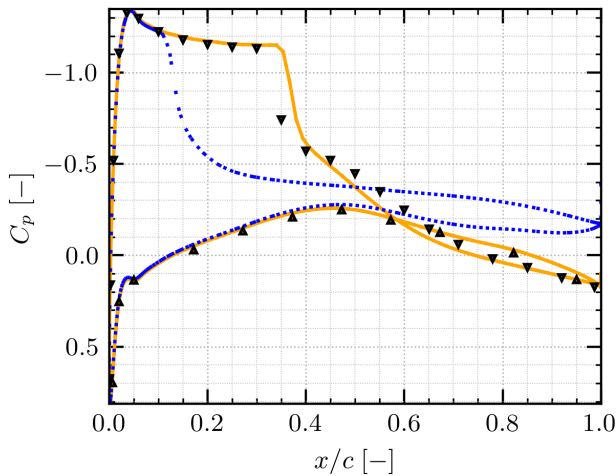
During the early development of CODA's Reynolds-stress model (RSM) for entry into service, I produced the first CODA RSM results for significantly separated flows using the ONERA M6 test case. The results below compare the SST and RSM models, demonstrating the RSM model's drastic improvement in accuracy for separated flow with respect to experimental data, owing to its completely different separation pattern.



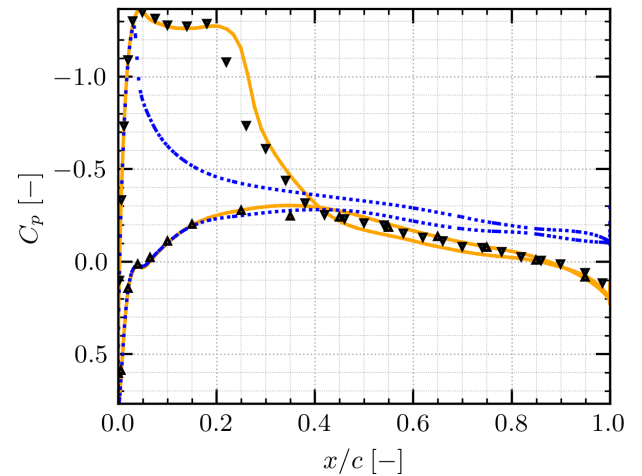
RSM streamlines at Mach = 0.84, $\alpha = 3^\circ$.



SST streamlines at Mach = 0.84, $\alpha = 3^\circ$.



RSM (orange), SST (blue) comparison at $\eta = 65\%$.

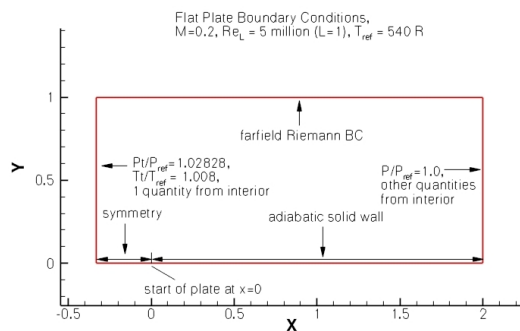


RSM (orange), SST (blue) comparison at $\eta = 80\%$.

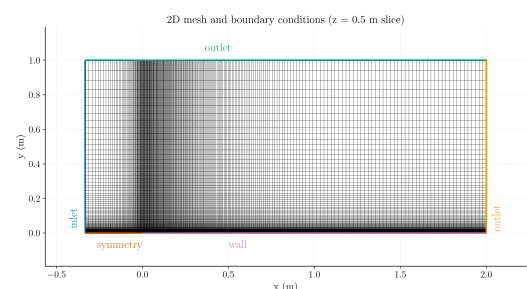
E2 Flat Plate Boundary Layer

Another open-source case I integrated into CODA's automated testing framework was the Flat Plate test case, used to assess boundary-layer results across turbulence models and CODA versions.

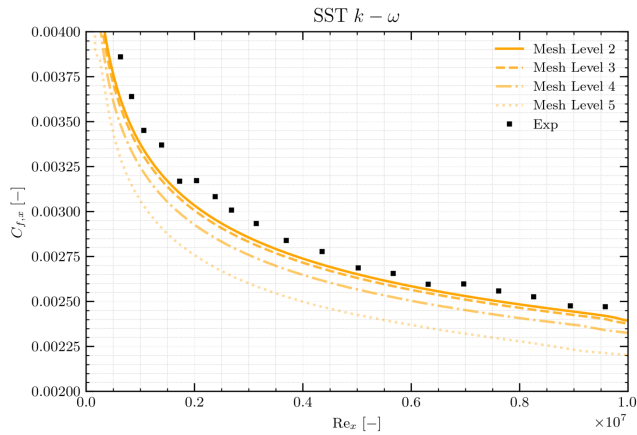
As part of the Airbus SST Task Force, I compared the ω and $\ln(\omega)$ formulations of CODA's SST model. This analysis examined their convergence behaviour to support selection of the formulation for entry into service.



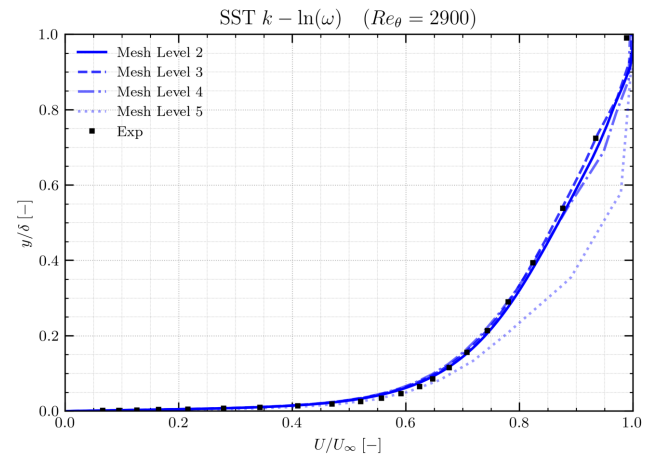
Flat Plate setup.



Flat Plate Mesh Level 3.



Skin friction profile convergence, ω formulation.



Inner BL profile convergence, $\ln(\omega)$ formulation.

E3 International Conferences

I travelled to Bremen, Germany, and Toulouse, France, to present key findings to more than 100 experts from Airbus, DLR and ONERA. In Bremen, I presented progress on the partner-enabled automated CODA testing framework. In Toulouse, I presented advances in CODA's RSM model in support of its entry into service.



Conference in Bremen, Germany.



Conference in Toulouse, France.

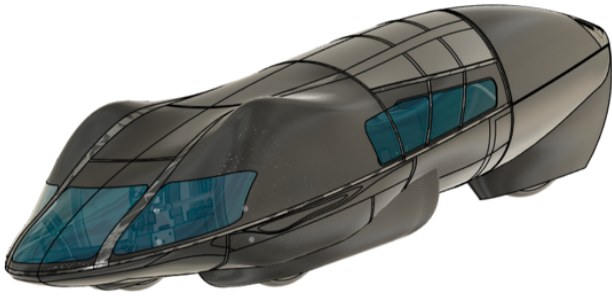
Imperial College London

MEng Aeronautical Engineering | 2025

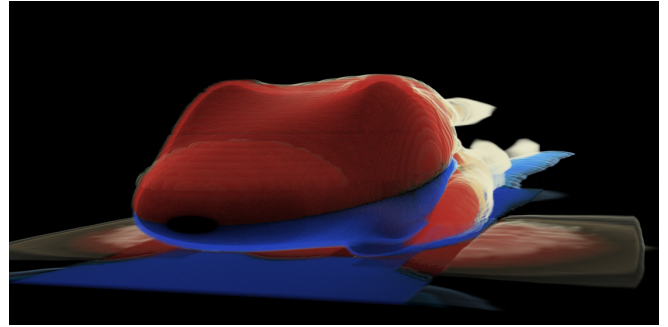
E4 Group Design Project Shell Eco-marathon Vehicle

Over two months, a 25 person team developed an upgrade package for Imperial's Shell Eco-marathon vehicle. I led the five person aerodynamics sub-team, using high-fidelity CFD to:

1. Validate the CFD model against an Imperial run half-scale wind-tunnel test using a digital-twin setup.
2. Assess the baseline design in a track simulation environment.
3. Iteratively surface and simulate design concepts to improve the vehicle's aerodynamic efficiency.



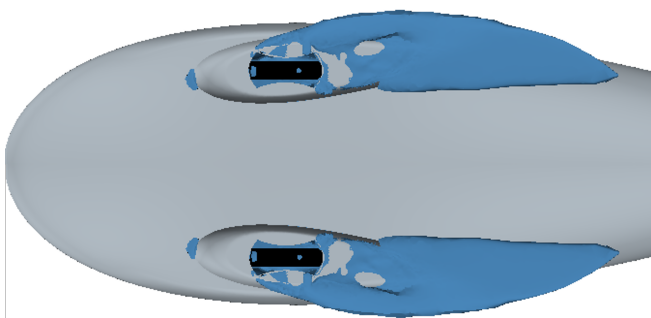
Detailed CAD model.



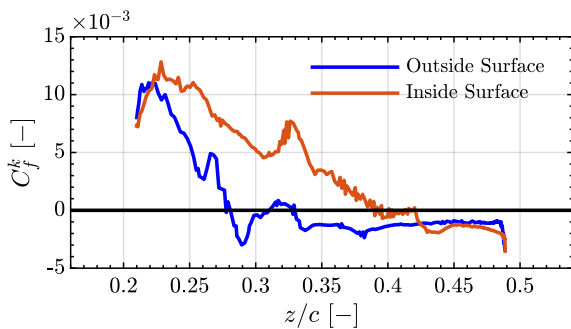
Colourful Fluid Dynamics.

E4.1 Aerodynamic Design

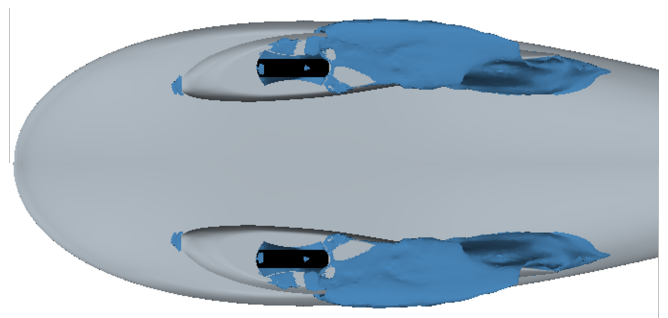
I surfaced iterative front wheel fairing designs to reduce the downstream separation volume. The figures below compare two airfoil section fairing iterations: geometry 770 reduces separation relative to geometry 776, as shown by the separation isosurfaces and skin-friction plots.



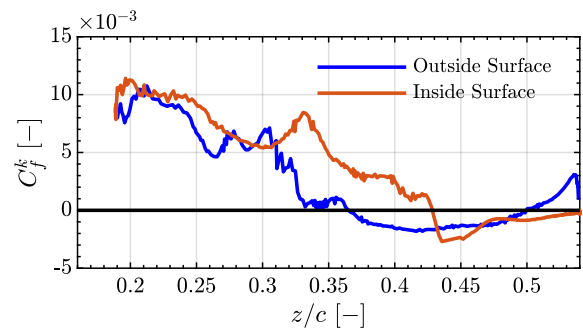
Separation isosurface for fairing ID 776.



Skin-friction for fairing ID 776.



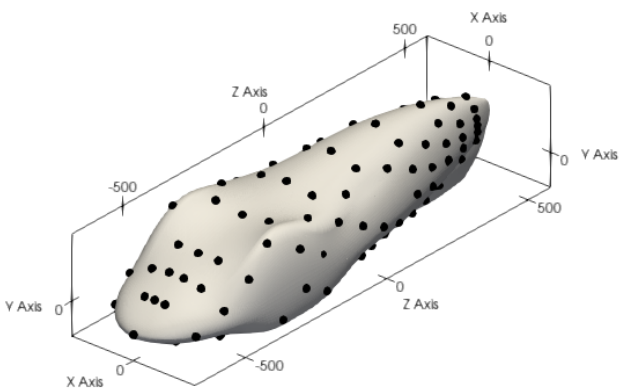
Separation isosurface for fairing ID 770.



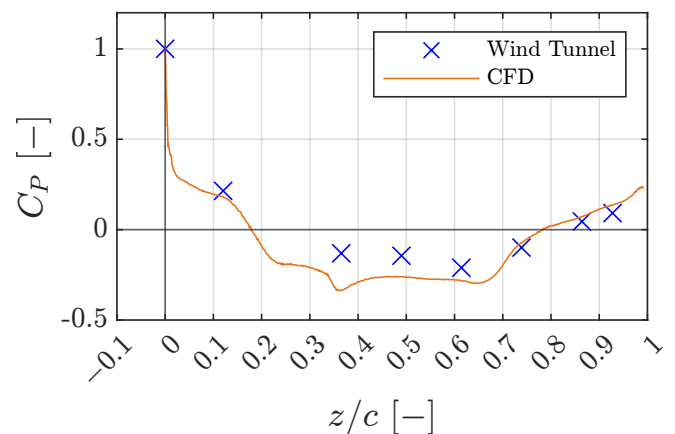
Skin-friction for fairing ID 770.

E4.2 Wind-tunnel Validation

This validation used an Imperial wind-tunnel test of the half-scale vehicle model, instrumented with pressure taps across axisymmetric planes. The digital-twin validation was highly successful with very strong agreement. The figure below shows the plane with the weakest agreement, where disagreement is confined to the downstream front-wheel separation region. All other regions on the car had a high accuracy with respect to the experimental data.



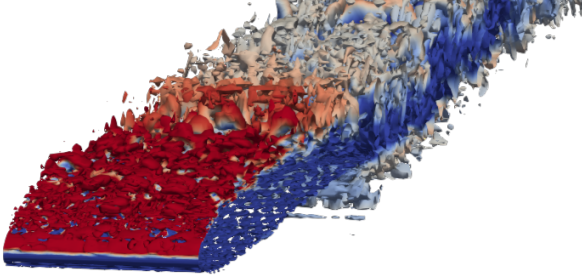
Pressure tap positions on wind tunnel model.



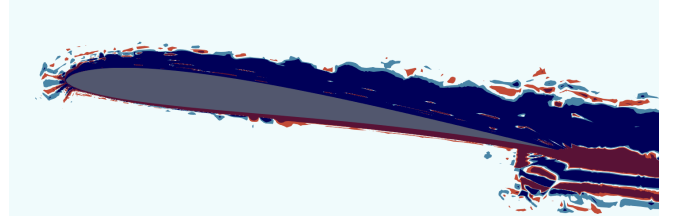
The lower surface vertical plane has the weakest agreement.

E5 Direct Numerical Simulation of an Airfoil

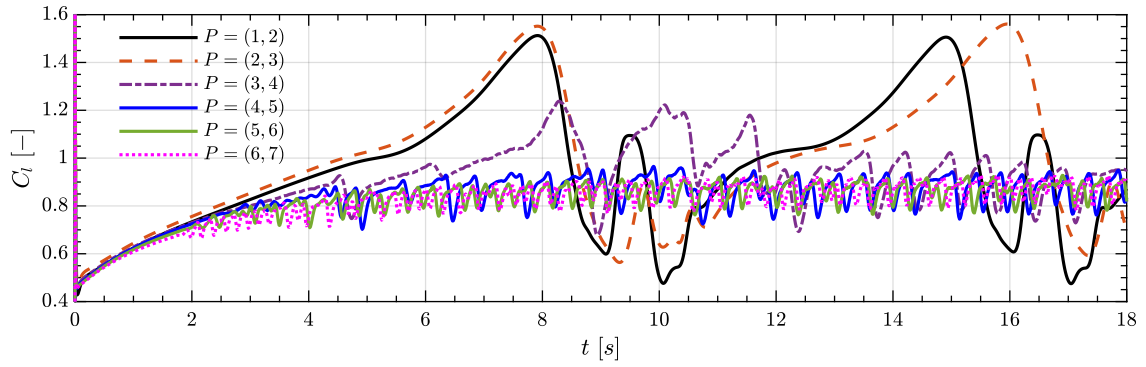
This four-month project was my first experience of scale-resolved simulation. I used the spectral/hp solver Nektar++ to perform direct numerical simulations of unsteady, transitional flow over the AG16 airfoil. I also ran steady RANS simulations in STAR-CCM+ and used XFOIL, enabling comparison between the different approaches.



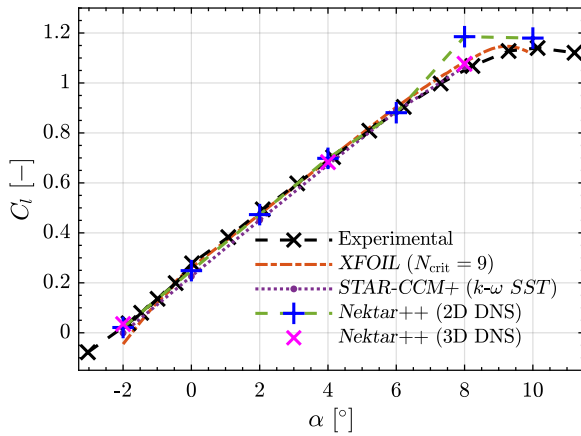
3D Q-criterion snapshot.



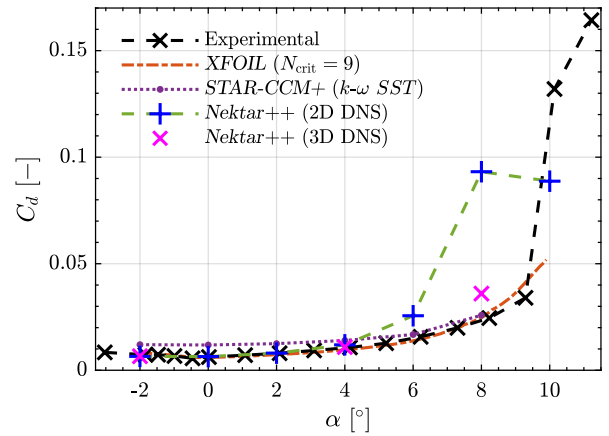
Spanwise averaged vorticity snapshot, LE separation.



Unsteady polynomial order convergence for $\alpha = 6^\circ$, $\text{Re} = 300032$.



Lift-coefficient comparison.



Drag-coefficient comparison.